Sector Gated Graph Neural Networks for Company Bankruptcy Risk

Project report | RiskAssesment | 3 November 2026

Abstract

The project explores bankruptcy ranking based on company characteristics, legal information about lawsuits between companies and persons and a graph of companies and persons. Target sector is an industry. A learned gate retains its companies and related context, and chooses extra exterior nodes and weights typed messages. In addition to neighborhood averaging, there are explicit relation counts. Controlled variants include numerical embeddings, more complex aggregation, aggregate vectors, sector-conditioned blending and pairwise ranking loss.

The ensemble validation selected from the recorded ensemble has test ROC AUC 0.8170, a company bootstrap 95% interval of 0.7775-0.8530 and average precision 0.8902 on 491 test records. The embedding, aggregation and ranking variant is an improvement over the reference by 0.0337 in terms of AUC. These results do not provide calibrated probabilities and do not form the basis for a prospective prediction or performance. Overlap of observation time and outcome-aligned observations limit interpretation.

1 Introduction

1.1 Application and purpose

It is a research system to rank bankruptcy risks of small and medium-sized enterprises in an industry of interest. It can aid an analyst identify businesses to evaluate more thoroughly, as well as include relationships via supervisor(s), executives, stakeholders, investors, and various other businesses. This is a model and a benchmarking pipeline for the implementation of Python; this repository does not provide a production lending platform or operational user interface.

Financial-ratio approaches [1, 2] motivate firm-level assessment, while graph methods motivate using related entities. SMEsD supplies business attributes and legal-event information rather than complete accounting ratios. The project combines these inputs with observed relationship structure and asks which external context helps a sector-specific task.

1.2 Scope and objectives

Financial-ratio approaches [1, 2] inspire the evaluation at the level of the firm, and the graph methods inspire the use of related entities. SMEsD does not provide fully fledged accounting ratios, only business attributes and legal-event information. The project is bringing together these inputs and observed relationship structure and enquires about what external context aids a sector-specific task? The goals are: to keep and score all companies in a chosen sector; to understand the importance of external nodes and edges; to keep the frequency of the relationship between the nodes, which averaging could throw away; to be able to compare architecture extensions by a common protocol; and to report discrimination with uncertainty. The positive class is bankruptcy while the negative class is survival.

They are dealing with the retrospective classification on a Chinese-SME dataset. It doesn't include verified forecasting from a common historical cutoff, bankruptcy-time estimation, recommendations and proven generalization to other countries. The contribution is an implemented sector conditioned set of known learning ideas that are backed up by ablation evidence and not claiming that each component is new.

2 Related Work

These next 15 articles cover the topics of bankruptcy modeling, table learning, graphs and the features used for this project. Gaps are specific to the project assessment, not to statements from original authors about their inability to achieve these. There are no conclusions drawn for comparison with results from other datasets.

Paper	Contribution	Research gap and project relevance
[1] Altman (1968) Financial ratios and discriminant analysis	Uses accounting ratios in a multivariate discriminant model to distinguish between failed and surviving firms. The basic bankruptcy strategy of the company.	Fails to capture relationship between company and individuals or the context in the target industry. SMES cannot reproduce themselves every 24 hours either.
[2] Ohlson (1980) Probabilistic bankruptcy prediction	Applies conditional logit modeling to appear in relation to company financial features and bankruptcy as a probabilistic substitute to discriminant analysis.	Does not capture relational dependency explicitly. Questions for evaluation that remain: Transportability and calibration. Promotes binary supervision and the distinction between scores and actual probabilities.
[3] Chen and Guestrin (2016) XGBoost	Introduces scalable regularized tree boosting, sparse-data handling and efficient approximate tree construction.	Tabular summarization must occur instead of message passing in relations. Connects a matched tree baseline with motivation. The captured GBDT baseline does not have to be XGBoost.
[4] Kipf and Welling (2017) Graph convolutional networks	Proposes an efficient neighborhood propagation method based on node attributes and graph structure that is used in semi-supervised classification.	Homogeneous convolution does not capture the selection of industry conditioned context and business relation family differentiation explicitly. Weights and explicit counts are incorporated into the project.
[5] Hamilton et al. (2017) GraphSAGE	Knows how to apply neighborhood sampling and aggregation functions to create representations for nodes they haven't seen before.	There are no “must select” or “must encode for” constraints for generic aggregation. Promotes bounded context processing despite the fact that this project learns gates and linked context.

2 Related Work continued

Paper	Contribution	Research gap and project relevance
[6] Velickovic et al. (2018) Graph attention networks	Learns different attention weights for neighbors and aggregates their transformed features.	Attention alone does not produce sector node selection or the number of relations in the sector name/relation name bankruptions. Motivates learned message importance, final variants have gated averages instead of the optional attention layer.
[7] Schlichtkrull et al. (2018) Relational graph convolution	Extends graph convolution to multi-relational data and evaluates entity classification and link prediction.	Knowledge graph results are not specific to the target sector or budgeting for the context of bankruptcy. Motivates families of relations, the project contains common message transformations as well as scale factors per relation.
[8] Wang et al. (2019) Heterogeneous graph attention	Combines node-level and semantic-level attention over meta-path neighborhoods.	Meta-path aggregation does not include the target industry gate mechanism. Motivates persons, not just companies, as the active loader does so for heterogeneous learning.
[9] Feng et al. (2019) Hypergraph neural networks	Uses hyperedge convolution to encode higher-order associations among groups of entities.	Groups by themselves do not solve typed pairs or validity over time. Motivates industry, geographical, and qualification groups; the optional project HyperLayer is deactivated in final variants..
[10] Xu et al. (2019) How powerful are GNNs	Analyzes expressive limitations of neighborhood aggregation and develops the graph isomorphism network.	Expressivity does not yield empirical bankruptcy accuracy or protection against sampling bias. Motivates maintaining multiplicity via counts; the implementation is not a GIN.

2 Related Work continued

Paper	Contribution	Research gap and project relevance
[11] Xu et al. (2018) Jumping Knowledge networks	Combines representations at different neighborhood depths to accommodate local graph structure.	Not responsible for selecting which industries from outside are relevant or if additional depth is beneficial to SMEsD. The project combines the outputs of the layers and assesses this alongside aggregation and degree alterations.
[12] Corso et al. (2020) Principal neighbourhood aggregation	Combines multiple aggregators and degree scalers to represent continuous-feature neighborhoods more effectively.	General benchmarks for graph performance do not provide optimal bankruptcy aggregation method. Inspires weighted average, max and degree-scaled average; the project is PNA-derived, but not a complete replication.
[13] Gorishniy et al. (2022) Numerical feature embeddings	Studies piecewise-linear and periodic encodings for numerical inputs to tabular neural networks.	Table advantages may not transfer to graph choice when faced with distribution shifts. The project selects eight training-quantile bins and four embedding sizes per column; TE validates this addition.
[14] Burges et al. (2005) RankNet	Introduces differentiable probabilistic pairwise learning of ranking functions.	Search ranking results do not validate bankruptcy adjustment or cross-entropy improvement. Inspires positive-negative logistic pair loss; TR assesses it separately while TEAR incorporates it among other modifications.
[15] Wei et al. (2024) ComRisk and SMEsD	Combines business and lawsuit information with heterogeneous and hypergraph encoders and releases SMEsD.	Not tested for this sector-specific gating mechanism. Provides the most relevant prior benchmark; an aligned leakage-free evaluation is required due to input and methodology differences

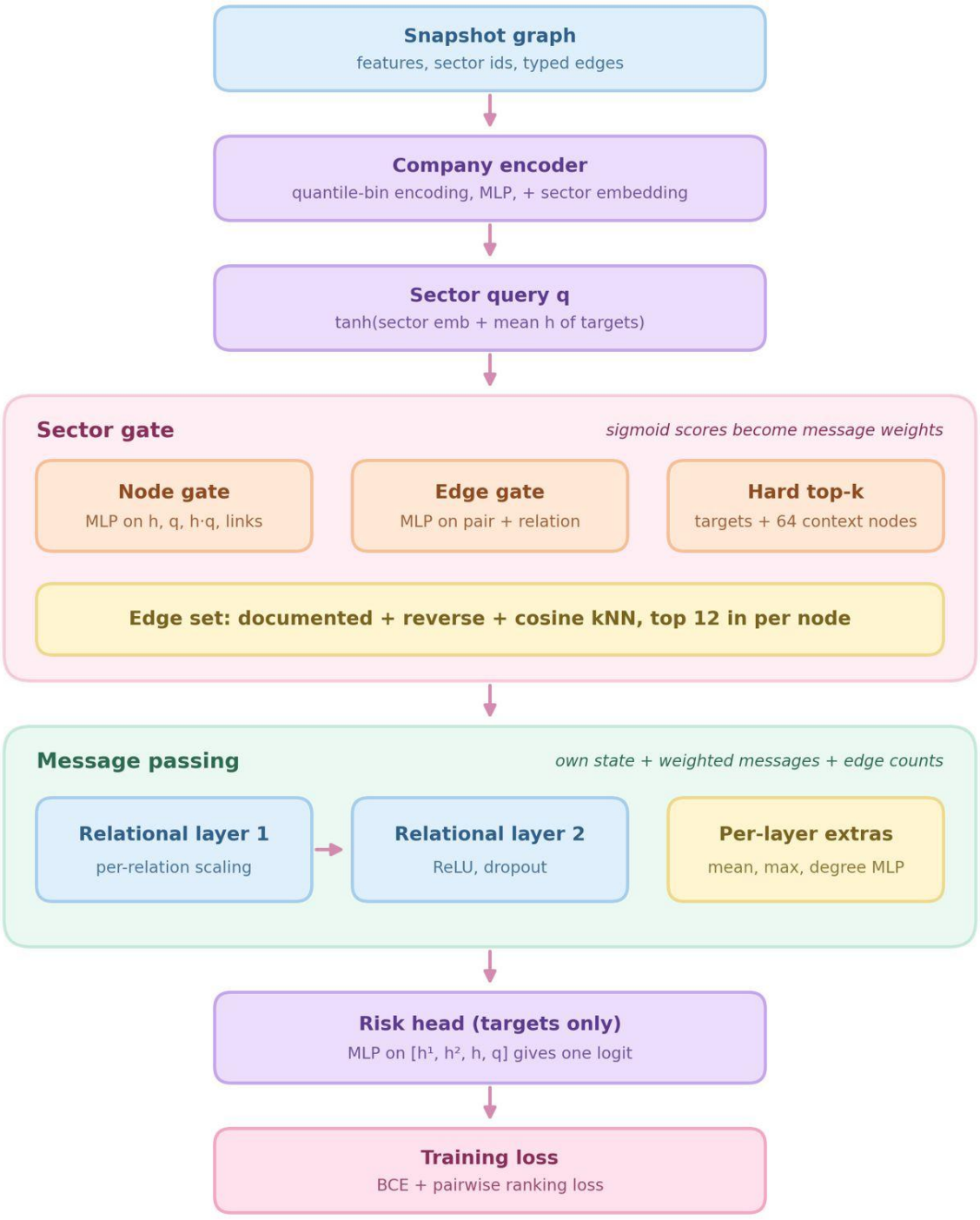
2.1 Research gap addressed

The project introduces the concept of selecting contexts from the industry, propagating typed company-person relations, and explicitly counting the number of persons in a neighbourhood from one bankruptcy ranking pipeline. However, its experiments will test combinations that are useful, but fail to isolate the gate against all alternatives. To prove the independence of the value of the gate, matched no-gate, fixed-gate, random-gate and degree-only controls must be provided on a new temporal holdout.

3 Methodology

3.1 Problem formulation and dataset

For each released split, let G = (V, E, X, R) contain companies and persons, typed edges, node features and relation families. Let T_s contain the companies assigned to industry s. The model produces one logit z_i for each i in T_s. External companies and persons provide context but do not receive target labels in that sector pass. A sigmoid transforms logits to model scores; probability interpretation requires calibration.



Item	Description
Released split records	Training 2,816; validation 721; test 491; total 4,028 records across splits.
Company identities	The upstream dataset describes 3,976 company identities. The 4,028 split records must not be described as 4,028 distinct companies.
Positive prevalence	Training about 57.6%; validation 49.1%; test 64.8%.
Features	Registered and paid in capital, set up age, lawsuit category counts or learned lawsuit vectors.
Graph and sectors	Seven families of models, 20 types of raw edges. The industry defines sectors are also available, as are area and qualification groups.
Documented overlap	There are 35 shared company IDs for train-validation and 17 for train-test inspections, documented in project inspection notes.
Evidence available	Code and Summary CSVs are provided in the package as well as no raw SMES data or per-company predictions.

Counts, overlap and feature audits are documented in DATASET_NOTES.md and the loader. The upstream split periods are 2014-2018, 2019 and 2020-2021. The paper lists company counts of 2,848/741/505, while its label counts align with the smaller released split sizes. This discrepancy matters when comparing published scores [15; dataset repository].

3.2 Observation time and temporal validity

The project notes explain the concept of age, relative to an observation time explained by the project's outcome alignment: the bankruptcy time for failed firms and December 2021 for survivors. Thus, there is no common prediction cutoff established from the data. The as_of date of the loader, 31 December 2021, isn't evidence that all the inputs were available before the output.

All round four variants remove recent-lawsuit count, newest-lawsuit age and mean-lawsuit age. Set-up age (except T0NA). Pre-processing using vector preprocessing in a lawsuit still keeps at least 48 records when sorting by the age of the lawsuit. This means that the encoder doesn't specify when things happened, but for long histories, selection may be based on age. Timing-free: Pertaining to the input columns that have been removed, not whether it forecasts without leakage.

3 Methodology continued

3.3 End to end flow diagram

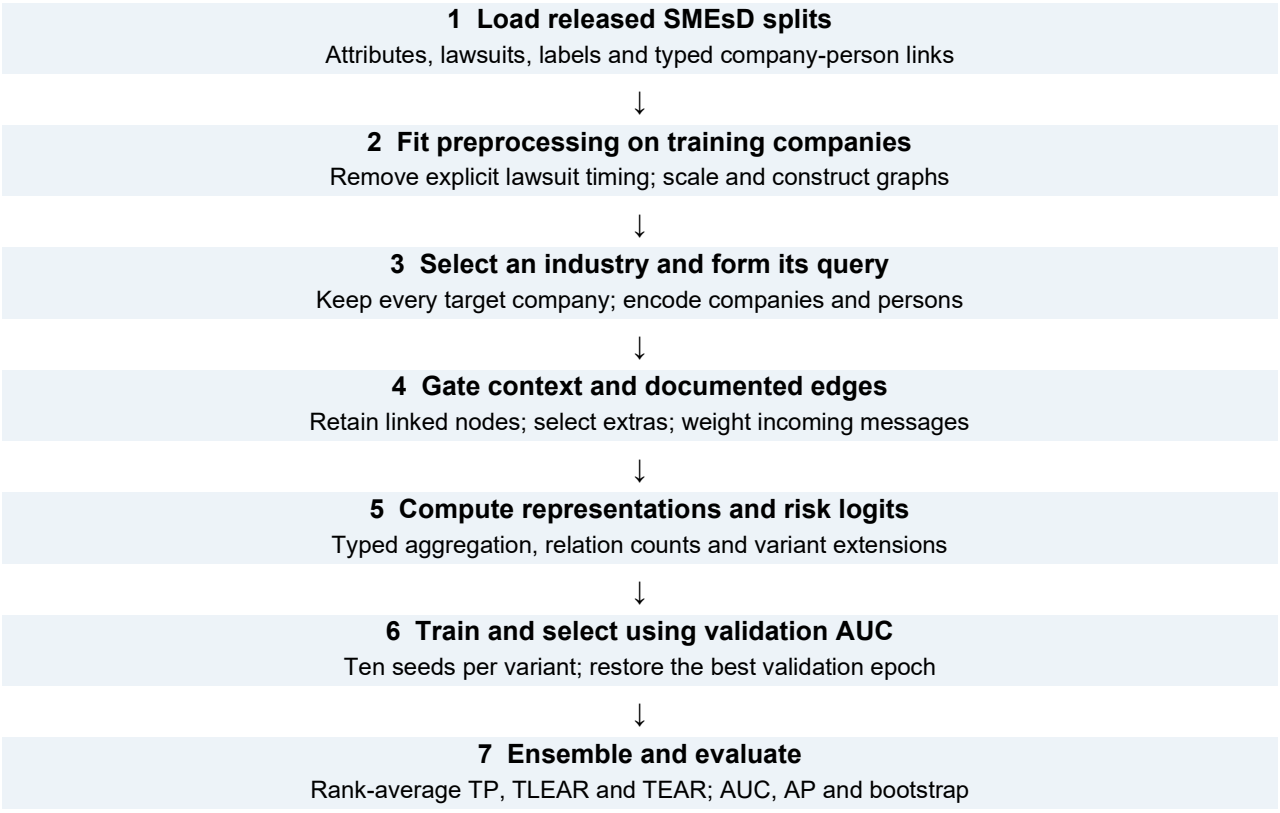


Figure 1. Main steps of the implemented workflow. Validation selects epochs and the headline. Earlier test-shift diagnostics nevertheless influenced the feature protocol.

3.4 Feature construction

Capital values are clipped below at zero and transformed with log(1 + value); set-up age is expressed in years. Lawsuit summaries include counts over 11 causes, four court categories and four outcomes, total count and availability. Removing three timing columns from the original 27-dimensional company vector leaves 24 features. RobustScaler is fitted only on training companies; transformed features are clipped to [-5, 5].

Snapshots add is_person column. Other persons begin with no attributes and a dedicated sector embedding; therefore, there will be no upstream pretrained embeddings for round four. The lawsuit-vector variants keep just three business attributes (static) and add learned legal-event representations. Industry terms are gathered across splits, in a transductive schema selection that should be reconsidered for totally new industries.

3 Methodology continued

3.5 Heterogeneous graph and sector gate

Explicit company-person graphs are built, not the company-only projection (which is also provided by the utility code). The raw types transform into supervisor, executive, stakeholder and investor families, and investment by companies, branches and financial-links. Undirected family level triples are removed. The documentation of edges in both directions, same family identity, are used in round four.

With initial encoded state h_i and sector embedding e_s, the query is q_s = tanh(e_s + mean(h_i for i in T_s)). The node gate uses h_i, q_s, their elementwise product and typed evidence of direct links to the sector. All targets and directly linked outside nodes are retained, followed by up to 32 additional high-scoring nodes. TP also retains external companies reached through linked persons. The value 32 therefore limits extra context, not total retained context.

The edge gate scores source and destination states, the query, their elementwise product and relation embedding. It keeps up to 64 incoming edges per destination. The algorithm performs a non-linear activation (sigmoid function) on the edge score a_ji, followed by a multiplication by weights w_j and w_i for a selected edge j to i. Target weights are 1s, context weights are the scores of the sigmoid nodes. The top-k membership function is discrete, meaning that the gradients flow over the selected weights, not the selected indices.

3.6 Typed propagation and structural counts

A message is m_ji = W_msg h_j elementwise-multiplied by the learned relation scale r_ji. Its weighted mean is M_i = sum(a_ji m_ji) / max(sum(a_ji), epsilon). The base layer computes h_i_next = ReLU(W_self h_i + M_i + D_i). D_i is a learned transform of log(1 + c_ir), where c_ir counts selected incoming edges of relation r. These counts describe the retained graph; full-graph degree features are not appended in round four.

The richer block is adding the maximum weighted message and weighted mean times log(1 + total incoming weight). A degree MLP gets logs and presents of relation counts. The summary shorthand sum/max/mean is hence a weighted mean and not a raw-sum aggregator, maximum and degree-amplified mean.

3.7 Sector blend and other optional layers

TP employs two layers and a sector-conditioned blend at persons that are in the first layer. In-sector and external company messages are summarized in the separated weighted and scaled by learned relation-specific sigmoid gates. An out-of-sector contribution penalty of 0.1 times the mean external gate discourages out-of-sector contributions to excessive levels. Gate values represent model behaviours, not causal pathways to risk. The final round-four variants implement the optional hypergraph and relational-attention layers, but they do not employ them.

3 Methodology continued

3.8 Numerical and lawsuit representations

TE, TEAR and TLEAR use piecewise-linear numerical embeddings [13]. Eight quantile bins per input column are fitted on training companies. Values are represented as fractions of progression through bins (with a clipping range of [0, 1]) and each mapped to 4 learned dimensions per column. Embeddings are added to the original data. The embedding component saturates outside of its range, but the shifted input values are not lost in the network so it is not invariant to shifts.

TL and TLEAR store lawsuit cause, court and outcome in 6, 3 and 3 dimensions, respectively. The MLP generates an 16-dimensional event vector and stores events in its attention. A flag indicating whether a lawsuit has been filed is added, as are a retained-record count and other log data. The age of an event is not used and learned recency decay/recent flags are disabled. Histories are safely masked if they are missing. There is still the usual maximum of 48 events to preprocess.

3.9 Training objective and configuration

Binary cross-entropy with logits is the core loss. Ranking variants add L_rank = mean(log(1 + exp(z_negative - z_positive))) across bankrupt-surviving pairs in a batch [14]. The coefficient is 1.0; the loss is zero for a single-class batch. Sector-blend regularization is added when enabled. Batches combine six sectors and normalize loss by their target-company count.

Setting	Round four configuration
Width and depth	32 hidden units; one layer in T0; two in T2, TP, TA, TEAR and TLEAR.
Optimization	AdamW; learning rate 0.003; weight decay 0.0001; cosine annealing.
Regularization	Dropout 0.1; gradient-norm clipping at 5.0.
Training duration	Maximum 120 epochs; patience 20; validation-AUC improvement threshold 0.0001.
Repeated fits	10 seeds numbered 42-51; CPU with two threads by default.
Selection	Restore best validation-AUC checkpoint; average logits over seeds within each variant.

3.10 Evaluation and headline selection

ROC AUC gives half credit for ties when ranking positive/negative. Average precision is a summary of the precision vs. recall incremental and is prevalence dependent. The no-skill AP reference is about 0.648 at 64.8% positive labels and random-ranking ROC AUC is 0.5. Both measurements are without decision threshold or calibration verification.

The report script ranks variants according to the validation AUC and takes the average of the top three variants and then selects the best validation candidate that contains that ensemble. T0NA is a diagnostic only. It includes 3,000 company level bootstrap resamples, percentile intervals and shared draws for paired AUC contrasts. These intervals incorporate uncertainty from trained predictions but not uncertainty from retraining and/or selection. Depending on the companies being linked together in the graph, clustered resampling might be necessary.

4 Results and Discussion

4.1 Recorded results

The summary, rounded up to four decimals, is reproduced in Table 3. The ensembles are ten-seed logit ensembles and ENS3 comprises three ensembles by percentile rank. These are the outputs of recorded experiments, NOT a new training run for this report. The package does not include any raw predictions to re-create the intervals.

Variant	Valid AUC	Test AUC	95% AUC interval	Test AP
T0	0.8520	0.7898	0.7485 - 0.8299	0.8713
T2	0.8639	0.7990	0.7577 - 0.8374	0.8754
TP	0.8736	0.7981	0.7564 - 0.8371	0.8732
TL	0.8540	0.7956	0.7548 - 0.8344	0.8805
TE	0.8501	0.7870	0.7448 - 0.8261	0.8707
TA	0.8623	0.8162	0.7776 - 0.8530	0.8891
TR	0.8550	0.7900	0.7488 - 0.8293	0.8716
TEAR	0.8643	0.8236	0.7846 - 0.8590	0.8968
TLEAR	0.8722	0.8168	0.7769 - 0.8533	0.8877
T0NA	0.8422	0.7735	0.7298 - 0.8152	0.8516
ENS3	0.8758	0.8170	0.7775 - 0.8530	0.8902

Table 3. Source: report4_models.csv. T0 is the reference; T2 adds depth; TP adds two-hop context and sector blend; TL adds lawsuit vectors; TE adds numerical embeddings; TA adds richer aggregation, degree MLP and layer-history concatenation; TR adds ranking loss. TEAR combines E, A and R; TLEAR also uses lawsuit vectors. T0NA drops set-up age.

4.2 Headline and generalization

The main variants with the three highest AUCs for validation are TP, TLEAR and TEAR. ENS3's validation AUC is 0.8758, test AUC 0.8170, test AP 0.8902 and is the headline. However, TEAR has the largest individual-variant test AUC, 0.8236, but using such a test to replace the headline would involve outcome of the test being used for selection purposes.

The difference in the AUC between validation and test sets is ~0.0588 for ENS3. The following are not mechanical explanations of an AUC drop but they do have an impact on AP. In the absence of being able to partition the aggregate outputs into separate contributors, possible contributors that are plausible and could not be separated include temporal distribution shift, sector composition, small-sample variability, and the selection of validation.

If the 17 test companies which are also in the training data set are excluded, the ENS3 AUC changes from 0.8170 to 0.8160. This is not a useful sensitivity check, but rather retraining without overlap. Does not cover features for train-validation overlap, graph dependency, or outcome-aligned features. The ENS3 is a rank score and is not meant to be interpreted as a probability of bankruptcy.

4 Results and Discussion continued

4.3 Ablation results

Paired contrast	AUC change	95% bootstrap interval
Depth: T2 - T0	+0.0091	-0.0037 to +0.0218
Embedding: TE - T0	-0.0029	-0.0131 to +0.0080
Aggregation bundle: TA - T2	+0.0173	+0.0049 to +0.0310
Ranking: TR - T0	+0.0002	-0.0061 to +0.0064
Context and blend: TP - T0	+0.0083	-0.0085 to +0.0254
Lawsuit vectors: TL - T0	+0.0058	-0.0088 to +0.0215
Combined model: TEAR - T0	+0.0337	+0.0145 to +0.0535
Add lawsuits: TLEAR - TEAR	-0.0068	-0.0207 to +0.0061
Remove age: T0NA - T0	-0.0164	-0.0313 to -0.0012

Table 4. Source: report4_contrasts.csv. Intervals use paired company resamples. No adjustment for multiple contrasts is recorded.

Richer summaries. TA improves over the depth-matched T2 by 0.0173, with interval 0.0049 to 0.0310. With Means, multiplicity is not in the mix and strong individual messages can be concealed. These differences are hinted at by counts, presence indicators and maxima, and several neighborhood depths are left behind in layer-history concatenation. The result is for the combined block not the individual effects of each addition.

Isolated additions. The estimate for T2 is small and positive, but has an interval that contains zero. TE is a bit poorer and TR is substantially the same. It is not enough that supervision is more flexible or ranks oriented. The improvement of TEAR over T0 is 0.0337 (interval 0.0145 to 0.0535), but this combines contrast cannot prove a specific interaction between the interactions of the components.

Lawsuits and external context. Small positive estimates with intervals that cross zero are found for TL and TP. The addition of vectors from lawsuits leads to a reduction of the estimation by 0.0068, with an interval that crosses zero as well. It is possible that limited lawsuit coverage and small sample size explain this, rather than these factors being causal.The possibility of limited lawsuit coverage and the small sample size is a potential explanation for this, but not a causal explanation.

Age and degree. The AUC decreases by 0.0164 when set-up age is removed. Predictive validity does not imply temporal validity. The project inspection notes also show a degree-augmented gradient-boosted model, with an AUC of 0.800 compared to 0.631 (without degree). Here these are historical checks that do not stand in pairs with a matched tree baseline and are more of a contextual evidence of the past.

4 Results and Discussion continued

4.4 Comparison with published work

Table 5 uses SMEsD values reported in Wei et al. [15], Table III. This is a reference comparison across protocols, not a paired test. The other reviewed papers supply methodological foundations; their scores on unrelated tasks cannot establish relative bankruptcy performance.

Published system	Published AUC	ENS3 minus published
GBDT	0.6843	+0.1327
GCN	0.7099	+0.1071
GAT	0.6251	+0.1919
RGCN	0.6857	+0.1313
HAN	0.7331	+0.0839
HGNN	0.6433	+0.1737
ie-HGCN	0.7560	+0.0610
Loss-weighted ComRisk	0.8256	-0.0086
ComRisk	0.8483	-0.0313

Table 5. Published values from [15]; ENS3 AUC 0.8170. Original implementations were not rerun for this report.

The ENS3 outperforms all other mentioned traditional and general GNN-based baselines, but is inferior to ComRisk by 0.0313 and loss-weighted ComRisk by 0.0086. Differing split counts, relation processing, lawsuit timing, representation data, and checkpoints preclude attributing any differences solely to architecture. There is no evidence for which protocol or architecture makes ComRisk more accurate.

4.5 Limitations and future work

The reporting code shows that the timing columns have been removed post-test shift diagnoses. The validation set chosen for round four does not undo any test-reuse that has already occurred. There must be an entirely new hold-out period. The feature and linking filtering must be performed based on the time prior to the prediction period.

These benchmarks must also include the comparison with the company-only model, degree augmented tree, no edge network, degree preserving shuffled graph and gate fixed/random benchmarks. Such comparisons will help us determine whether the identity of the counterparty is informative beyond the mere counts. We must include the metrics at the sector level, calibration, Brier score, and precision recall on thresholds as well. The stored summaries cannot support invented accuracy, F1, ROC curves, runtime, or fairness results.

4.6 Conclusion

The example shows sector-specific bankruptcy ranking where the AUC 0.8170 has been achieved. The most explicit proof of the architecture design is provided by the richer aggregate and structural-count block. The next step to be taken is that of independent evaluation using proper observation times and structural controls.

5 References

All fifteen reviewed papers are listed with full authorship, publication venue and a DOI or author-preprint identifier. Links lead to bibliographic records or full text.

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6 Code and Dataset

6.1 GitHub links and accessibility

Project code - sector-gated RiskAssesment implementation
https://github.com/ananyashriram10/RiskAssesment

Dataset - SMEsD files in the original authors repository
https://github.com/shaopengw/ComRisk/tree/main/data

Dataset documentation
https://github.com/shaopengw/ComRisk/blob/main/SMEsD.md

Original ComRisk code and dataset attribution
https://github.com/shaopengw/ComRisk

On 3 November 2026, the project page was publicly readable, and returned HTTP 200 without authentication. These are independent repositories: RiskAssesment provides the sector-gated model and ComRisk provides original data and comparator. A public access does not constitute a redistribution license. No data license is identified on the project notes or on inspection and the data is linked but not redistributed.

6.2 Source organization and evidence

Source location	Role
README.md; docs/DATASET_NOTES.md	Application, AUDIT dataset, History and limits.
risk_gnn/model.py; risk_gnn/data.py	Graph Schema, gates, typed layers, embeddings and risk head.
benchmark/data.py; model_ext.py	The pre-processing, relation mapping and lawsuit encoder.
benchmark/run2.py; run4.py	Common configuration, variants, losses and training protocol.
benchmark/report4.py	Ensemble selection, Bootstrap and Overlap analysis.
results/summary/report4_models.csv; report4_contrasts.csv	Numerical evidence given for Tables 3 and 4.
tests/; kaggle/	Specify the content and skills of unit tests and keep a benchmark notebook.

This report provides a description of the supplied snapshot of the local source. The Summary CSVs are validated against the reporting logic and against the README. Raw data and predictions/training logs are missing and therefore scores were not independently reproduced. The supplied runtime is missing PyTorch and pytest; there is no new test pass. Requirements do not guarantee an environment, they only specify minimum versions.

6 Code and Dataset continued

6.3 Reproduction procedure

Run from a project checkout in an isolated Python environment. Shell scripts require Bash, such as Git Bash, WSL or Linux. The fetch script clones ComRisk into the location expected by the loader.

git clone https://github.com/ananyashriram10/RiskAssesment.git

cd RiskAssesment

python -m venv .venv

Activate .venv for your shell.

pip install -r requirements.txt

bash scripts/fetch_data.sh

python -m pytest -q

Bash: main variants and age diagnostic

for v in T0 T2 TP TL TE TA TR TEAR TLEAR T0NA; do

  python -m benchmark.run4 --variant "$v" --tag "r4_$v" --seeds 10

done

python -m benchmark.report4

The trainer writes test and validation prediction CSVs and metadata JSON under results/. The report script reads the report and recognizes the best validation variants, builds ENS3 and prints report4_models.csv and report4_contrasts.csv in results/. This project contains historical summaries, which have been provided in the results/summary/ directory. There are no three-variant headlines that can be reproduced using only one variant.

6.4 Reproducibility record and future validation

An experimental release should contain the project commit, the upstream data commit, the versions of installed packages, arguments, best epochs for the seed, raw predictions and report tables. The fetch script does not fetch the revision that is identified by the notes in the datasets, but rather the current revision of the repository. Actually revise and check intended data set snapshot.

If you're interested in following up on the study, keep the current benchmark as exploratory evidence to serve as a new entity-disjoint chronological holdout, and freeze selection rules ahead of time. Introduce graph-dependence-aware uncertainty, calibration and matched structural controls. This makes it possible to tell whether an experiment was repeated or whether the saved experiment is evidence of future bankruptcy forecasting of a new population.